

AP Precalculus

Unit 3: Trigonometric and Polar Functions

Lesson 3.6 Sinusoidal Function Transformations

Name: _____ Date: _____

Period: _____

Learning Targets — I Can...

1. I can identify the amplitude, midline (vertical shift), period, and phase shift of a sinusoidal function from its equation.
(LO 3.6.A)
2. I can explain how each parameter a , b , c , d in $f(\theta) = a \sin(b(\theta + c)) + d$ corresponds to a specific additive or multiplicative transformation of the parent sine function.
(LO 3.6.A)
3. I can rewrite a cosine function as a phase-shifted sine function and explain why sine and cosine are equivalent up to a horizontal translation.
(LO 3.6.A, EK 3.6.A.1)

What's in This Packet

#	Component	Description	Score
1	Warm-Up	Spiral review: unit-circle values, parent sine/cosine graphs, transformation vocabulary	_____
2	Guided Notes	Parameters a , b , c , d ; amplitude, period, phase shift, midline; sine–cosine equivalence	_____
3	Group Activity	Identify and interpret sinusoidal parameters in context (tiered R/A/E)	_____
4	Exit Ticket	Identify all four parameters from a given equation; match equation to verbal description	_____
5	Homework	Mixed: parameter identification, period/amplitude computation, AP-style MC	_____
Total			_____